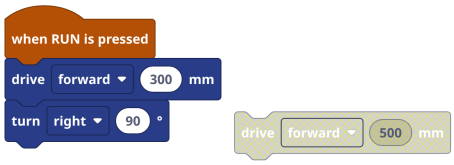
Blockly tips and tricks

The basics, variables, and the handy things nobody finds by looking.

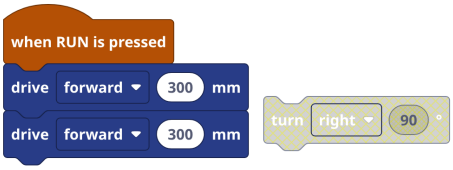
Click a block to pick it. **Right-click** it for its menu. A **gear** changes its shape, a **triangle** is a list to choose from, a **?** is a note. Cards marked HIDDEN are tricks you only find if someone shows you.

1 Snap it on, or it does nothing



Only blocks under **when RUN is pressed** (or in a subroutine) run. A loose block turns **gray** and is skipped.

2 Pull out just one block HIDDEN



A drag takes every block under it too. Hold **Ctrl** while you drag to take just that one. Or click it, press **X**.

3 Duplicate HIDDEN



Right-click, **Duplicate**: a copy of the block and all inside it. Faster: click it, press **D**.

4 Keys HIDDEN

`Ctrl+C`	`Ctrl+V`	copy, paste (other tabs too)
`Ctrl+X`		cut
`Ctrl+Z`	`Ctrl+Y`	undo, redo
`Delete`		delete it
`D`		duplicate it
`X`		pull it out of its stack
`C`		clean up: line blocks up

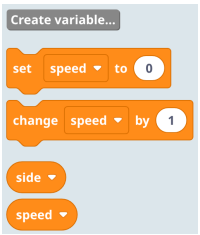
Click a block first. Mac: **Cmd** for Ctrl.

5 Numbers are slots HIDDEN



Click a number to type. Drop a **variable** or a **math** block on it. Drag that out: the number comes back.

6 Make a variable



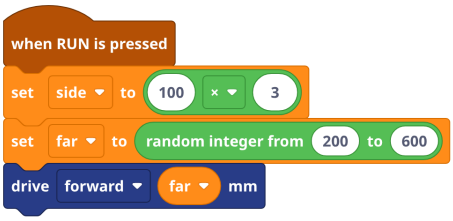
A box with a name, for a number or some text. **Variables**, **Create variable...**, type a name like **side**.

7 Start it with a number



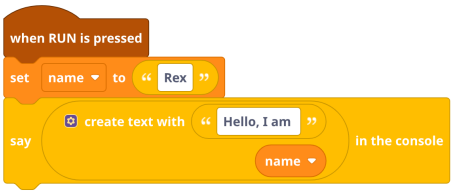
set side to 300, then put the round **side** where a number goes. Change 300 once: every move follows.

8 Start it with math



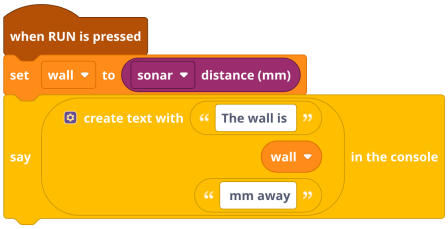
Drop a **Math** block in: **100 × 3**, or **random integer 200 to 600** for a new number every run.

9 Start it with text



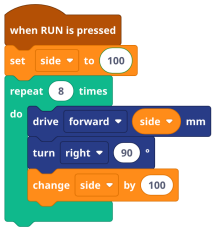
The **" "** block (**Text**) holds words. **create text with** glues words and numbers together for **say**.

10 Start it from a sensor



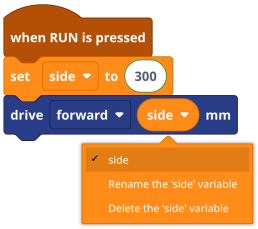
set wall to sonar distance keeps what the sonar saw right then. Read it again later: did it change?

11 Change it as you go



change side by 100 adds 100 each time round: a spiral. **-100** makes it smaller.

12 Rename or delete it HIDDEN



Click the **triangle** on a variable: pick another one, **Rename** it everywhere, or **Delete** it.

Subroutines, gears and menus

Make your own blocks. Then the menus that hide behind a right-click.

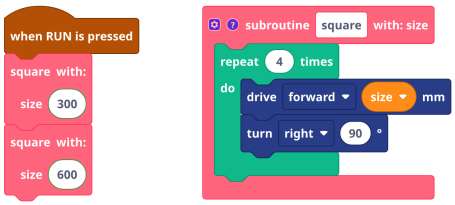
What does it do? Rest the mouse on a block: a hint pops up.

Set it first. A variable never set is empty: **set** it at the top.

Watch it. say a variable each time round, simulator at **real time**.

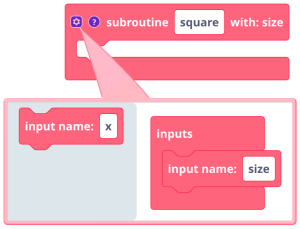
Keep it. Programs live in this browser. **Share / phone...** saves a file.

13 Make a subroutine



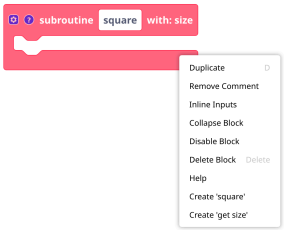
Drag **subroutine** out of **Subroutines**, click its name to rename it, fill it. Its call (**square**) is in **Subroutines** now.

14 The gear adds inputs HIDDEN



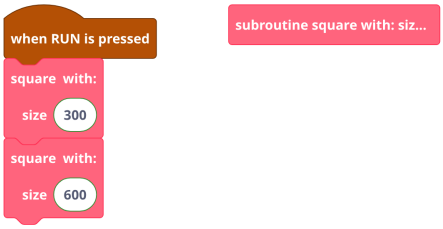
Click the **gear**. Drag **input name** into **inputs**, click the **x**, type **size**. Click the gear again to close.

15 Use the input inside HIDDEN



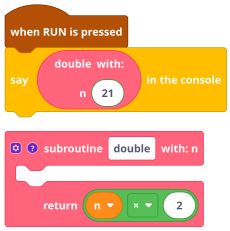
An input is a variable. **Right-click** the subroutine: **Create 'get size'** makes its block, **Create 'square'** a call.

16 Give it a value



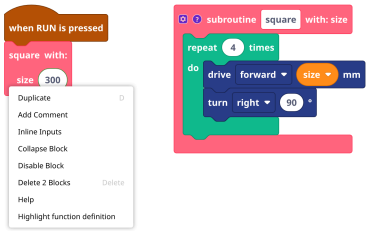
Each call now has a slot: **square with size 300**, then **600**. One subroutine, any size.

17 A subroutine that answers HIDDEN



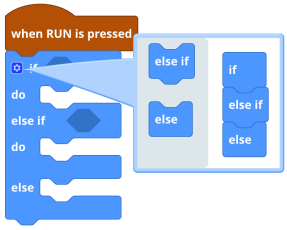
The second **subroutine** block ends in **return**: it gives back a number, so its call fits in a slot.

18 Where is that subroutine? HIDDEN



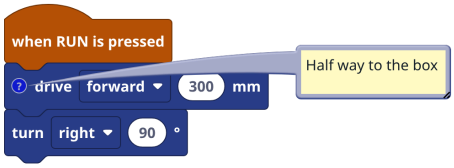
Right-click a call: **Highlight function definition** shows the subroutine it runs.

19 Gears on if and text HIDDEN



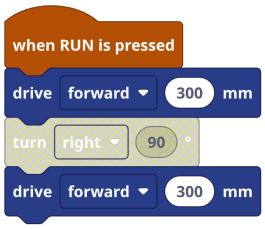
On **if**: drag in **else if** and **else**. On **create text with** and **create list with**: more **items**.

20 Notes for people HIDDEN



Right-click, **Add Comment**. Click the **?** to read it. The program ignores it. Subroutines come with one.

21 Turn a block off HIDDEN



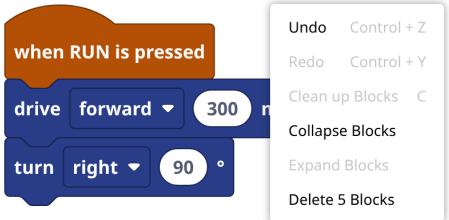
Right-click, **Disable Block**: it is skipped. Run it with and without. **Enable Block** turns it back on.

22 Fold it up HIDDEN



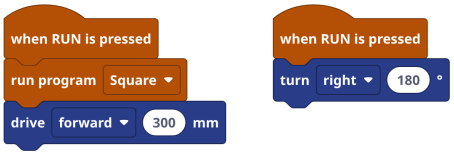
Right-click, **Collapse Block**: one short line. It still runs. **Expand Block** opens it.

23 Right-click empty space HIDDEN



Clean up Blocks lines them up. **Collapse** or **Expand** all. **Delete** all (Ctrl+Z brings them back).

24 Run another program HIDDEN



run program (**Start**) runs another tab, then goes on. Two **when RUN** blocks: the top one runs first.